

CBSE Class 12 Maths — Relations and Functions & Inverse Trigonometric Functions

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark (MCQ/Assertion-Reason/VSA)

- Q1.** Write the principal value of $\cos^{-1}(\cos \frac{7\pi}{6})$. [PYQ 2023] [Confidence: Very High]
- Q2.** Find the principal value of $\sin^{-1}(\sin \frac{2\pi}{3})$. [PYQ 2022] [Confidence: Very High]
- Q3.** Let R be the relation in the set $A = \{1, 2, 3\}$ given by $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}$. State whether the relation R is reflexive, symmetric, or transitive. [PYQ 2020] [Confidence: High]
- Q4.** Let the function $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 3x - 4$. Is the function f bijective? [PYQ 2019] [Confidence: High]
- Q5.** Find the domain of the inverse trigonometric function $f(x) = \cos^{-1}(2x - 1)$. [Practice] [Confidence: Medium]
- Q6.** Write the principal value of $\tan^{-1}(-\sqrt{3})$. [PYQ 2018] [Confidence: Very High]
- Q7.** Let $A = \{1, 2, 3\}$. Find the number of equivalence relations containing $(1, 2)$. [PYQ 2021] [Confidence: Medium]
- Q8.** Write the range of the principal value branch of $y = \sec^{-1} x$. [Practice] [Confidence: High]
- Q9.** Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^2$. Is this function one-one? [PYQ 2019] [Confidence: High]
- Q10.** What is the principal value of $\cot^{-1}(-1)$? [PYQ 2017] [Confidence: Very High]
- Q11.** Find the principal value of $\sec^{-1}(-2)$. [PYQ 2018] [Confidence: High]
- Q12.** Let R be a relation on the set of natural numbers $\mathbb{N}$ defined by the equation $x + 2y = 8$. Write the domain of the relation R . [PYQ 2020] [Confidence: Medium]
- Q13.** Evaluate: $\sin(\frac{\pi}{3} - \sin^{-1}(-\frac{1}{2}))$. [PYQ 2023] [Confidence: Very High]
- Q14.** Show whether the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = 2x$ is an onto function. [Practice] [Confidence: High]
- Q15.** Evaluate the value of $\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2)$. [PYQ 2024] [Confidence: Very High]
- Q16.** Find the value of $\sin(\tan^{-1} x + \cot^{-1} x)$, where $x \in \mathbb{R}$. [PYQ 2016] [Confidence: High]
- Q17. Assertion (A):** The function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = |x|$ is not one-one. **Reason (R):** $f(1) = f(-1) = 1$, but $1 \neq -1$. Choose the correct option. [Practice] [Confidence: Very High]
- Q18. Assertion (A):** The domain of the function $y = \sin^{-1} x$ is $[-1, 1]$. **Reason (R):** The principal value branch of $\sin^{-1} x$ has the range $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Choose the correct option. [PYQ 2024] [Confidence: High]
- Q19.** Let R be a relation in the set of real numbers $\mathbb{R}$ defined as $R = \{(a, b) : a \leq b\}$. Is the relation symmetric? [PYQ 2017] [Confidence: High]

Q20. Determine if $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = [x]$ (greatest integer function) is an onto function. [Practice] [Confidence: Medium]

2-Mark

Q21. Check whether the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{x^2+1}$ is one-one, onto, or bijective. [PYQ 2022] [Confidence: High]

Q22. Evaluate: $\tan(\sin^{-1}(\frac{3}{5}) + \cot^{-1}(\frac{3}{2}))$. [PYQ 2020] [Confidence: High]

Q23. Express $\tan^{-1} \left[\frac{\sqrt{1+x^2}-1}{x} \right]$ in the simplest form, where $x \neq 0$. [PYQ 2018] [Confidence: Medium]

Q24. Prove that $2 \sin^{-1}(\frac{3}{5}) = \tan^{-1}(\frac{24}{7})$. [PYQ 2019] [Confidence: Very High]

Q25. Prove that the relation R defined on the set of integers $\mathbb{Z}$ as $R = \{(a, b) : a - b \text{ is an integer}\}$ is an equivalence relation. [Practice] [Confidence: High]

Q26. Solve the following equation for x : $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$. [PYQ 2017] [Confidence: Medium]

Q27. Find the exact principal value of $\cos^{-1}(-\frac{1}{2}) + 2 \sin^{-1}(\frac{1}{2})$. [PYQ 2021] [Confidence: Very High]

Q28. Let $A = \{1, 2, 3\}$. Check whether the relation $R = \{(1, 2), (2, 1)\}$ is transitive or not. Justify your answer. [PYQ 2020] [Confidence: High]

Q29. Express $\tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right)$ in the simplest form, where $0 < x < \pi$. [PYQ 2019] [Confidence: Very High]

Q30. Prove that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 1 + x^2$ is neither one-one nor onto. [Practice] [Confidence: High]

Q31. Find the principal value of $\sin^{-1}(\sin \frac{3\pi}{5})$. [PYQ 2023] [Confidence: Very High]

Q32. Let R be a relation in the set of all lines in a plane defined by $L_1 R L_2 \iff L_1$ is perpendicular to L_2 . Show that R is symmetric but neither reflexive nor transitive. [PYQ 2018] [Confidence: High]

3-Mark

Q33. Show that the relation R in the set of integers $\mathbb{Z}$ given by $R = \{(a, b) : 2 \text{ divides } (a - b)\}$ is an equivalence relation. [PYQ 2019] [Confidence: Very High]

Q34. Let $A = \mathbb{R} - \{3\}$ and $B = \mathbb{R} - \{1\}$. Consider the function $f : A \rightarrow B$ defined by $f(x) = \frac{x-2}{x-3}$. Show that f is a bijective function. [PYQ 2020] [Confidence: High]

Q35. Evaluate: $\tan^{-1}(1) + \cos^{-1}(-\frac{1}{2}) + \sin^{-1}(-\frac{1}{2})$. [Practice] [Confidence: High]

Q36. Prove that $\cot^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right) = \frac{x}{2}$, for $x \in (0, \frac{\pi}{4})$. [PYQ 2018] [Confidence: Medium]

Q37. Prove that $\tan^{-1}(\frac{1}{2}) + \tan^{-1}(\frac{1}{5}) + \tan^{-1}(\frac{1}{8}) = \frac{\pi}{4}$. [PYQ 2016] [Confidence: Medium]

Q38. Solve for x : $\sin^{-1}(1 - x) - 2 \sin^{-1} x = \frac{\pi}{2}$. [PYQ 2022] [Confidence: High]

Q39. Show that the relation R on the set $A = \{1, 2, 3, 4, 5\}$ given by $R = \{(a, b) : |a - b| \text{ is even}\}$ is an equivalence relation. [PYQ 2019] [Confidence: Very High]

Q40. Find the value of $\tan\left(\frac{1}{2}\cos^{-1}\frac{\sqrt{5}}{3}\right)$. [Practice] [Confidence: High]

Q41. Prove that the function $f : \mathbb{N} \rightarrow \mathbb{Y}$ defined by $f(n) = n^2$, where $\mathbb{Y}$ is the range of f , is an invertible function. [PYQ 2017] [Confidence: Medium]

Q42. Show that the relation R in the set A of all points in a plane given by $R = \{(P, Q) : \text{distance of } P \text{ from origin} = \text{distance of } Q \text{ from origin}\}$ is an equivalence relation. [PYQ 2023] [Confidence: Very High]

5-Mark (Long Answer/Case-Study)

Q43. Show that the relation R in the set $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$ given by $R = \{(a, b) : |a - b| \text{ is a multiple of } 4\}$ is an equivalence relation. Find the set of all elements related to 1. [PYQ 2020] [Confidence: Very High]

Q44. Prove that the function $f : \mathbb{R} \rightarrow \{x \in \mathbb{R} : -1 < x < 1\}$ defined by $f(x) = \frac{x}{1+|x|}$, for all $x \in \mathbb{R}$, is both one-one and onto. [PYQ 2019] [Confidence: Very High]

Q45. Let $A = \{1, 2, 3, \dots, 9\}$. Let R be the relation on $A \times A$ defined by $(a, b)R(c, d)$ if and only if $a + d = b + c$. Prove that R is an equivalence relation. [PYQ 2022] [Confidence: High]

Q46. Let $f : [0, \infty) \rightarrow [-5, \infty)$ be defined by $f(x) = 9x^2 + 6x - 5$. Show that f is an invertible function and find $f^{-1}(y)$. [PYQ 2018] [Confidence: Medium]

Q47. Show that the relation R in the set $A = \{1, 2, 3, 4, 5\}$ given by $R = \{(a, b) : |a - b| \text{ is divisible by } 2\}$ is an equivalence relation. Also, write all the equivalence classes of R . [PYQ 2021] [Confidence: High]

Q48. Show that $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{4x+3}{6x-4}$ (where $x \neq \frac{2}{3}$) is bijective. Additionally, find the inverse of f . [PYQ 2017] [Confidence: Very High]

Q49. Case Study: Students of Class 12 are organizing a group activity. They map out relations among themselves. Let A be the set of all students in the class. A relation R_1 is defined as $R_1 = \{(a, b) : a \text{ and } b \text{ are of the exact same height}\}$. (i) Prove that R_1 is an equivalence relation. (3 Marks) (ii) If another relation $R_2 = \{(a, b) : a \text{ is strictly taller than } b\}$ is defined, check if R_2 is symmetric and transitive. (2 Marks) [Practice] [Confidence: High]

Q50. Case Study: Let A be the set of all textbooks in a school library. The librarian defines a relation $R = \{(x, y) : x \text{ and } y \text{ have the same number of pages}\}$. (i) Show that R is an equivalence relation. (3 Marks) (ii) Can this relation R be considered a bijective function if mapped from book to book? Explain why or why not. (2 Marks) [Practice] [Confidence: Medium]

Answer Key

Q1. Principal value branch of $\cos^{-1} x$ is $[0, \pi]$. $\cos(7\pi/6) = \cos(2\pi - 5\pi/6) = \cos(5\pi/6)$. Hence, $\cos^{-1}(\cos(5\pi/6)) = 5\pi/6$. **Q2.** $\sin^{-1}(\sin \frac{2\pi}{3}) = \sin^{-1}(\sin(\pi - \frac{\pi}{3})) = \sin^{-1}(\sin \frac{\pi}{3}) = \frac{\pi}{3}$. **Q3.** It contains $(1, 1), (2, 2), (3, 3)$ so it is reflexive. It contains $(1, 2)$ but not $(2, 1)$, so not symmetric. It contains $(1, 2)$ and $(2, 3)$ but not $(1, 3)$, so not transitive. Reflexive only. **Q4.** Yes, it is bijective. It is linear, $f(x_1) = f(x_2) \implies x_1 = x_2$ (one-one), and $y = 3x - 4 \implies x = (y + 4)/3 \in \mathbb{R}$ (onto). **Q5.** For domain of $\cos^{-1}(2x - 1)$, we need

$-1 \leq 2x - 1 \leq 1 \implies 0 \leq 2x \leq 2 \implies 0 \leq x \leq 1$. Domain is $[1]$. **Q6.** $\tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}$.

Q7. The equivalence relations are the universal relation $A \times A$ and

$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$. So, 2 relations. **Q8.** Range is $[0, \pi] - \{\frac{\pi}{2}\}$. **Q9.** No.

$f(1) = 1$ and $f(-1) = 1$. Two inputs map to the same output. **Q10.**

$\cot^{-1}(-1) = \pi - \cot^{-1}(1) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$. **Q11.** $\sec^{-1}(-2) = \pi - \sec^{-1}(2) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$.

Q12. $x, y \in \mathbb{N}$. If $y = 1, x = 6$. If $y = 2, x = 4$. If $y = 3, x = 2$. If $y = 4, x = 0 \notin \mathbb{N}$. Domain (x values) = $\{2, 4, 6\}$. **Q13.** $\sin(\frac{\pi}{3} - (-\frac{\pi}{6})) = \sin(\frac{\pi}{3} + \frac{\pi}{6}) = \sin(\frac{\pi}{2}) = 1$. **Q14.** No. There is no pre-image for odd numbers like $y = 3$ in $\mathbb{N}$ (since $x = 3/2 \notin \mathbb{N}$). **Q15.** $\frac{\pi}{3} - \frac{2\pi}{3} = -\frac{\pi}{3}$. **Q16.**

$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$. So, $\sin(\frac{\pi}{2}) = 1$. **Q17.** Both Assertion and Reason are true, and R is the correct explanation of A. **Q18.** Both Assertion and Reason are true, but R is not the correct explanation of A. **Q19.** No. For example, $2 \leq 3$, so $(2, 3) \in R$, but $3 \not\leq 2$, so $(3, 2) \notin R$. **Q20.** No, the range of $f(x) = [x]$ is strictly integers $\mathbb{Z}$, not all real numbers $\mathbb{R}$.

Q21. Not one-one ($f(2) = 2/5, f(1/2) = 2/5$). Not onto (Range is $[-1/2, 1/2]$, which is not $\mathbb{R}$).

Neither. **Q22.** Let $\sin^{-1}(3/5) = A \implies \tan A = 3/4$. Let $\cot^{-1}(3/2) = B \implies \tan B = 2/3$.

Thus $\tan(A + B) = \frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{3}{4} \cdot \frac{2}{3}} = \frac{17/12}{6/12} = \frac{17}{6}$. **Q23.** Let $x = \tan \theta$. Expression becomes

$\tan^{-1}(\frac{\sec \theta - 1}{\tan \theta}) = \tan^{-1}(\frac{1 - \cos \theta}{\sin \theta}) = \tan^{-1}(\tan \frac{\theta}{2}) = \frac{1}{2} \tan^{-1} x$. **Q24.** Let

$\sin^{-1}(3/5) = \theta \implies \sin \theta = 3/5 \implies \tan \theta = 3/4$. LHS

$= 2\theta = \tan^{-1}(\frac{2 \tan \theta}{1 - \tan^2 \theta}) = \tan^{-1}(\frac{2(3/4)}{1 - 9/16}) = \tan^{-1}(\frac{24}{7})$. **Q25.** Reflexive: $a - a = 0 \in \mathbb{Z}$.

Symmetric: If $a - b \in \mathbb{Z}$, then $b - a = -(a - b) \in \mathbb{Z}$. Transitive: $(a - b) + (b - c) = a - c \in \mathbb{Z}$.

Q26. $\tan^{-1}(\frac{5x}{1-6x^2}) = \frac{\pi}{4} \implies 5x = 1 - 6x^2 \implies 6x^2 + 5x - 1 = 0 \implies x = 1/6$ or

$x = -1$. Since $x = -1$ gives a negative LHS, $x = 1/6$. **Q27.** $\frac{2\pi}{3} + 2(\frac{\pi}{6}) = \frac{2\pi}{3} + \frac{\pi}{3} = \pi$. **Q28.** Not

transitive. $(1, 2) \in R$ and $(2, 1) \in R$, but $(1, 1) \notin R$. **Q29.** Divide num and denom by $\cos x$:

$\tan^{-1}(\frac{1 - \tan x}{1 + \tan x}) = \tan^{-1}(\tan(\frac{\pi}{4} - x)) = \frac{\pi}{4} - x$. **Q30.** Not one-one: $f(1) = f(-1) = 2$. Not

onto: Range is $[1, \infty) \neq \mathbb{R}$ (e.g., no pre-image for 0). **Q31.** $\sin^{-1}(\sin(\pi - \frac{2\pi}{5})) = \frac{2\pi}{5}$. **Q32.**

Reflexive: A line cannot be perpendicular to itself (False). Symmetric: If $L_1 \perp L_2$, then $L_2 \perp L_1$

(True). Transitive: If $L_1 \perp L_2$ and $L_2 \perp L_3$, L_1 is parallel to L_3 , not perpendicular (False).

Q33. Reflexive: $a - a = 0$, divisible by 2. Symmetric: if $a - b = 2k, b - a = 2(-k)$. Transitive:

$(a - b) + (b - c) = 2k_1 + 2k_2 = 2(k_1 + k_2)$, so $a - c$ divisible by 2. **Q34.** One-one:

$\frac{x_1 - 2}{x_1 - 3} = \frac{x_2 - 2}{x_2 - 3} \implies x_1 x_2 - 3x_1 - 2x_2 + 6 = x_1 x_2 - 2x_1 - 3x_2 + 6 \implies x_1 = x_2$. Onto:

$y = \frac{x - 2}{x - 3} \implies x = \frac{3y - 2}{y - 1}$. For every $y \neq 1, x \in A$. **Q35.**

$\frac{\pi}{4} + \frac{2\pi}{3} + (-\frac{\pi}{6}) = \frac{3\pi + 8\pi - 2\pi}{12} = \frac{9\pi}{12} = \frac{3\pi}{4}$. **Q36.** Rationalize denominator or substitute

$1 \pm \sin x = (\cos(x/2) \pm \sin(x/2))^2$. Expression simplifies to $\cot^{-1}(\cot(x/2)) = x/2$. **Q37.**

$\tan^{-1}(1/2) + \tan^{-1}(1/5) = \tan^{-1}(\frac{7/10}{9/10}) = \tan^{-1}(7/9)$. Next,

$\tan^{-1}(7/9) + \tan^{-1}(1/8) = \tan^{-1}(\frac{7/9 + 1/8}{1 - 7/72}) = \tan^{-1}(\frac{65}{65}) = \frac{\pi}{4}$. **Q38.** Put $x = 0$, LHS

$= \sin^{-1}(1) - 0 = \pi/2$. (Using algebraic solution:

$1 - x = \sin(\pi/2 + 2 \sin^{-1} x) = \cos(2 \sin^{-1} x) = 1 - 2x^2 \implies 2x^2 - x = 0 \implies x = 0, 1/2$

. Checking $x = 1/2$ fails, so $x = 0$). **Q39.** Reflexive: $|a - a| = 0$ (even). Symmetric:

$|a - b| = |b - a|$. Transitive: $|a - b|$ even $\implies a, b$ same parity. $|b - c|$ even $\implies b, c$ same

parity. Thus a, c have same parity $\implies |a - c|$ is even. **Q40.** Let

$\cos^{-1}(\frac{\sqrt{5}}{3}) = 2\theta \implies \cos 2\theta = \frac{\sqrt{5}}{3}$. We need $\tan \theta$. $\tan \theta = \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}} = \sqrt{\frac{3 - \sqrt{5}}{3 + \sqrt{5}}} = \frac{3 - \sqrt{5}}{2}$.

Q41. Injectivity: $f(n_1) = f(n_2) \implies n_1^2 = n_2^2 \implies n_1 = n_2$ (since domain is $\mathbb{N}$). Surjectivity: by

definition of co-domain $\mathbb{Y}$ as range. Thus, invertible. $f^{-1}(y) = \sqrt{y}$. **Q42.** Reflexive: $OP = OP$. Symmetric: $OP = OQ \implies OQ = OP$. Transitive: $OP = OQ$ and $OQ = OR \implies OP = OR$.

Q43. Show reflexive, symmetric, transitive (similar to Q33 but multiple of 4). Equivalence class of 1: Elements $x \in A$ such that $|x - 1| = 4k$. Thus $x = 1, 5, 9$. Set = $\{1, 5, 9\}$. **Q44.** One-one: Let $f(x_1) = f(x_2)$. Two cases: both positive/both negative (yields $x_1 = x_2$) or opposite signs (leads to contradiction). Onto: Let $y \in (-1, 1)$. If $y \geq 0$, $x = \frac{y}{1-y} \geq 0$. If $y < 0$, $x = \frac{y}{1+y} < 0$. f is bijective.

Q45. Reflexive: $a + b = b + a \implies (a, b)R(a, b)$. Symmetric: $a + d = b + c \implies c + b = d + a \implies (c, d)R(a, b)$. Transitive: $(a, b)R(c, d)$ and $(c, d)R(e, f) \implies a + d = b + c$ and $c + f = d + e$. Add both: $a + d + c + f = b + c + d + e \implies a + f = b + e \implies (a, b)R(e, f)$. **Q46.** Write $f(x) = (3x + 1)^2 - 6$. Since $x \geq 0$, $3x + 1 \geq 1 \implies f(x)$ strictly increases (one-one). Range is $[-5, \infty)$, equal to codomain (onto). Inverse:

$y = (3x + 1)^2 - 6 \implies 3x + 1 = \sqrt{y + 6} \implies f^{-1}(y) = \frac{\sqrt{y+6}-1}{3}$. **Q47.** Proof of equivalence (same as Q39). Classes: Odd elements relate to each other, even to each other. Classes are $\{1, 3, 5\}$ and $\{2, 4\}$. **Q48.** One-one:

$\frac{4x_1+3}{6x_1-4} = \frac{4x_2+3}{6x_2-4} \implies -16x_1 + 18x_2 = -16x_2 + 18x_1 \implies 34x_1 = 34x_2 \implies x_1 = x_2$.
Onto: $y = \frac{4x+3}{6x-4} \implies 6xy - 4y = 4x + 3 \implies x(6y - 4) = 4y + 3 \implies x = \frac{4y+3}{6y-4}$. Inverse is $f^{-1}(y) = \frac{4y+3}{6y-4}$. **Q49.** (i) R_1 : Reflexive (a same height as a), Symmetric (a same as $b \implies b$ same as a), Transitive ($a = b$ and $b = c \implies a = c$). (ii) R_2 : Not symmetric (if a taller than b , b is NOT taller than a). Transitive (if $a > b$ and $b > c \implies a > c$). **Q50.** (i) Reflexive: book x has same pages as x . Symmetric: $x = y \implies y = x$. Transitive: $x = y$ and $y = z \implies x = z$. (ii) No, it cannot be a bijective function. A function requires each input to map to exactly one output. Here, one book maps to multiple books with the same page count, failing the "well-defined" property of a function.